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SHORT PATH AND SHORT CHAIN PROBLEMS IN THE PLANE

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ABSTRACT. In this paper, we revisit and explore variations of the shortest path problem within various obstacle settings. If the planar obstacles are disjoint it is natural to look for the shortest paths between two given points around impenetrable disjoint obstacles. If the obstacles form a covering of the plane then one could be interested in finding short paths traversing in the double covered regions of the plane. In computational geometry it is common to focus on the algebraic and computational hardness of the problem. In discrete geometry upper bounds for the lengths of the shortest path between two points are in the center of the study. In discrete geometry such problems were initiated by L. Fejes Tóth in the sixties and asked for a constant c such that the constructed paths' lengths are at most c times the distance between the starting and target points, as the distance approaches infinity. The purpose of this paper is twofold. We survey the main ideas, focusing on the intuition, behind those results which were initiated by questions of L. Fejes Tóth. We also provide a new short proof for the optimal constant in case of square packing. Concerning square coverings we give an upper bound for the length of the shortest path along the edges of the squares of the covering.

1. INTRODUCTION

The problem of computing shortest paths within a geometric domain has been well studied in computational geometry since the seventies. This type of research often focuses on the algebraic and computational hardness of computing the shortest path among obstacles and is relatively well studied; see the surveys [M97] and [M00].

Coincidentally, another branch of mathematics, discrete geometry, also considered shortest path problems. L. Fejes Tóth, founder of this field, popularised such questions dating back to the sixties. Fejes Tóth's questions were not about constructing the shortest paths between two points of distance d , he rather asked for upper bounds for the lengths of the shortest connecting paths within a family of obstacle fields. The purpose of this paper is to survey the main ideas behind those results which were initiated by questions of L. Fejes Tóth.

Initially even the terminology used by Fejes Tóth was different from that used in computational geometry.

Definition 1. We define a *layer* as a set of non-overlapping open regions lying within a parallel strip. Let w represent the width of the strip, and let ℓ denote the length of a path that connects the two edges of the strip without penetrating any of the regions. The *permeability* p of the layer is defined as

$$p = \frac{w}{\inf \ell}$$

Theorem 1 (L. Fejes Tóth [LFT66]). *A layer of width d consisting of congruent circles of radius 1 can always be crossed by a path of length $\ell \leq \frac{2\pi}{\sqrt{27}}(d - 2) + 2$.*

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Throughout this paper circles of radius 1 will be also called unit circles or unit discs (if we want to talk about the enclosed region). The above theorem states that the permeability of a layer of congruent unit circles is greater than $\frac{\sqrt{27}}{2\pi} = 0.82\dots$, which in the limit is the permeability for a sufficiently wide normal section of the densest lattice packing.

Shortly after L. Fejes Tóth's paper [LFT66] on layers consisting of circles, analogous results concerning other families of obstacles were proved (see surveys by A. Florian [F80a] and recently by G. Fejes Tóth and W. Kuperberg [GFT23]).

author/paper	disjoint obstacles	results on permeability
L. Fejes Tóth [LFT68]	congruent squares	Smallest permeability is equal to $\frac{2}{3}$.
L. Fejes Tóth [LFT68]	similar parallelograms	Smallest permeability is achieved with translates of parallelograms.
A. Florian H. Groemer [FG85]	homothetic copies of a regular m -gon ($m \geq 39$)	The smallest permeability among all packings cannot be reached using translates only.
B. Bollobás [B68]	similar parallelograms	Bollobás used a new approach and proved the result of L. Fejes Tóth [LFT68].
A. Florian [F78]	i) unit circles ii) translates of a discs of constant width	The smallest permeabilities in i) and ii) are equal to each other.
I. Hortobágyi [H76]	translates of a convex disc K with constant width	Among convex shapes the circle maximizes the smallest permeability (i.e. circle packings force one to make the longest path when traversing through a strip).
A. Florian [F79]	i) translates of a reg. hexagon ii) translates of a reg. triangle	The smallest permeabilities in i) and ii) are equal to $\frac{3}{4}$.
A. Florian [F80b]	i) translates of a convex disc K ii) translates of $K - K$	The smallest permeabilities in i) and ii) are equal to each other.

TABLE 1. Results on Permeability from 1966-1978

In addition to the specific permeability problems, L. Fejes Tóth realized that instead of asking about a short path across a layer of obstacles, it is natural to ask about a short path between any two given points where the obstacles are no longer confined to a layer. In fact, this general setup of two points, a packing or covering of obstacles, and a quantification of the length of a path will form the main foundation for the rest of the paper. The rest of the paper is organized in the following way.

- In Section 2 we state 4 basic short path problems. Each will be followed with a short history of solutions.

- In Section 3 we describe 5 basic strategies to find short paths in square and circle packings. At the heart of each paper, there is often a key elementary idea which makes a proof work. We took a closer look at 5 papers (L. Fejes Tóth's, J. Pach's, G. Fejes Tóth's, C. Papadimitreu's and M. Yannakakis', and A. Bezdek's) and describe their approaches in an intuitive manner. The five key ideas are different and hopefully inspire future research.
- In Section 4 we give a new, and short, proof of one of the problems detailed in Sections 2 and 3, namely finding short paths in packings of squares with side lengths at most one.
- In Section 5 we consider short paths in square coverings. This is a natural extension of Problem 3 of G. Fejes Tóth. We give new results, which are asking for further generalizations.

2. FOUR BASIC PROBLEMS

In this section we list four problems, asked by L. Fejes Tóth, G. Fejes Tóth, C. Papadimitriou, M. and Yannakakis, and A. Bezdek. Each problem is followed by the references of their solutions. We chose these problems because it is expected that they will inspire future results.

Traversing a layer of circles by a path of minimal length, as described in Section 1, is closely related to the problem of connecting two points with the shortest path. In fact, an upper bound L for the shortest path between points of distance w in the second version gives a lower bound $\frac{w}{L}$ for the permeability of a layer in a strip of width w . An implication in the other direction would be a harder question. This led L. Fejes Tóth to state the following research problem for circles:

Problem 1 (L. Fejes Tóth [LFT78]). *Consider a packing of disjoint circles of radii 1 in the plane. Prove that any two points of distance d , lying outside of the circles, can be connected by a path*

- (1) (strong version) of length $\leq \frac{2\pi}{\sqrt{27}}(d - 2) + \pi$
- (2) (weak version) of length $\leq \frac{2\pi}{\sqrt{27}}d + o(d)$

G. Fejes Tóth [GFT78] gave an existence proof for the strong version of the problem.

Problem 2 (L. Fejes Tóth). *Consider a packing of disjoint squares of edge length ≤ 1 in the plane. Prove that any two points of distance d , lying outside of the squares, can be connected by a path*

- (1) (strong version) of length $\ell \leq \frac{3}{2}d + \frac{1}{2}$
- (2) (weak version) of length $\ell \leq \frac{3}{2}d + o(d)$

The example on Fig. 1 (where vertical columns of unit squares are subsequently shifted down by $\frac{1}{2}$) shows that in Problem 2 the constant $\frac{3}{2}$ cannot be decreased. On the other end, it is easy to see that the following strategy achieves a path length $\leq 2d$. Starting at S , travel towards T until a square is hit. Travel around the boundary of the square back

to the line ST in whichever way is shorter. This heuristic is often known as the *return to the line* heuristic.

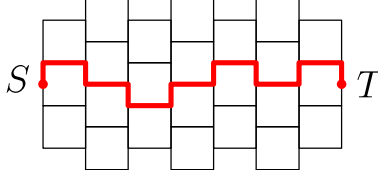


FIGURE 1. A squares packing and points S and T so that all connecting path have length at least $\frac{3}{2}|ST|$.

Using a mean value argument J. Pach [P77] gave an existence proof validating the weak version of the conjecture with $o(d) = \frac{3}{2} + 4\sqrt{d} + 1$. G. Fejes Tóth [GFT78] then gave an existence proof validating the strong version of the conjecture. C. Papadimitriou and M. Yannakakis [PY91] gave an algorithm to construct a path for the weak version of the conjecture with $o(d) = O(\sqrt{d})$. A. Bezdek [B99] gave an algorithm to construct a path which answers the weak version of the conjecture. The weak version will be revisited one more time in Section 4 in this paper, where the authors give a one page new proof with $o(d) = O(\sqrt{d})$ using their *evade a square and go straight towards the target heuristic*.

G. Fejes Tóth asked the following short path problem for coverings of the plane by various convex discs. In a sense this is the dual of the problem of L. Fejes Tóth concerning packings in the plane.

Problem 3 (G. Fejes Tóth). *Consider a covering of the plane by unit discs. Assume we have two points S and T at distance d apart which are contained in the doubly covered region of the covering. Prove that S and T can be connected by a path that is completely contained in the doubly covered region and has length $\leq \sqrt{2}d + O(1)$.*

When the centers of the unit circles form a unit square lattice and S and T are lattice points so that ST is parallel to diagonals of the squares, then it is easy to see that the connecting path has length $\sqrt{2}d$. A. Bezdek and D. Bagett [BB03] verified the above conjecture for lattice packings. Concerning the general circle covering, the upper bound of $\pi d + O(1)$ is easy to prove. E. Roldan-Pensado [P17] gave an upper bound of $(\frac{\pi}{3} + \sqrt{3})d + O(1) \approx 2.779d + O(1)$. Although this bound is still closer to πd than to $\sqrt{2}d$, it is considered to be an important improvement of this difficult problem.

Next we consider a new variant of short path problems. Let $\mathcal{F}$ be a locally finite covering of the plane by congruent copies of a specific centrally symmetric disc D . We say that two discs D, D' of the covering $\mathcal{F}$ are *connected by a chain of finitely many discs* $D_0, D_1, \dots, D_n$ of $\mathcal{F}$, if $D_0 = D$, $D_n = D'$ and consecutive discs D_i, D_{i+1} with $i = 0, \dots, n-1$ have a common point. The integer n will be called length of the chain. Given two of the discs in $\mathcal{F}$, we ask to find a connecting chain with as few discs as possible. More formally,

Problem 4 (A. Bezdek). *Consider locally finite coverings of the plane by unit squares. Prove that two squares of the covering whose centers are at distance d can be connected by a chain of squares of length $2\sqrt{2}d + o(d)$ and $2\sqrt{2}$ is the smallest possible constant in this bound.*

This problem was solved by A. Bezdek and L. Yuan in [BJ20]. The following square covering showed that $2\sqrt{2}$ cannot be replaced with a smaller constant.

Start with a unit square with a vertical diagonal, duplicate it and translate the second copy vertically down by a small distance ϵ (see the two squares in the circle on Fig. 2). Then take the union of these two squares and make two rows by tiling them according to Fig. 2. We will refer to these two rows as a single strip of squares. Finally, we use translates of this strip to tile the plane. Notice that any chain crossing this strip must contain at least four discs from this strip. Choose S and T far from each other so that the line ST passes through the strip as shown on Figure 2. Then taking ϵ sufficiently small we get that the shortest connecting chain has to have four squares along each segment of length $\simeq \sqrt{2} + 3\epsilon$. Thus, in the bound proved of Problem 4 the constant must be at least than $\frac{4}{\sqrt{2}}$.

A. Bezdek and L. Yuan also considered the same problem concerning coverings with circles of radius 1. They showed two circular discs of the covering whose centers are at distance d can be connected by a chain of discs of length $\frac{2}{\sqrt{3}}d + o(d)$. To decide if this bound is tight remains open. The case of circular discs is harder than that of squares. Among others, they presented a special circle covering, where sufficiently short chains have to include discs which do not intersect the segment connecting the centers of the first and last discs.

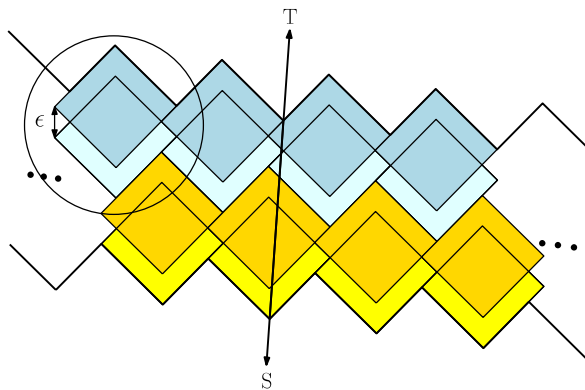


FIGURE 2. A square covering and points S and T so that all connecting chains have length $> 2\sqrt{2}|ST|$.

3. VARIOUS PROOF STRATEGIES TO FIND SHORT PATHS IN CIRCLE AND SQUARE PACKINGS

In this section we mention 5 theorems concerning upper bounds of short path lengths. Although the detailed proofs given by L. Fejes Tóth, J. Pach, G. Fejes Tóth, C. Papadimitrou, M. Yannakakis, A. Bezdek and O. Henderschedt are important, we want to focus on the author's approach of finding paths and estimating their length. We will explain the proof strategies in an intuitive manner and place them in text boxes for easy comparison.

3.1. Strategy 1: L. Fejes Tóth. According to Problem 1 L. Fejes Tóth had to show the existence of a short path crossing a layer of circles. He managed to prove the following.

Theorem 2 (L. Fejes Tóth [LFT78]). *Given any packing of circles in a layer of width w , there exists a path crossing the layer of length $\ell \leq \frac{2\pi}{\sqrt{27}}(w - 2) + 2$.*

How did L. Fejes Tóth construct short path to cross a layer of circles?

In his own words, L. Fejes Tóth wrote the following. ‘Imagine that the layer lies on an inclined plane so that the lines bounding the layer are horizontal. The circles are considered as inelastic low cylinders affixed to the plane. A particle dropped from a point U of the upper line will describe a path composed of straight segments and circular arcs of length $\leq \frac{\pi}{4}$ and will arrive at a point L of the lower line.’ Finally Fejes Tóth modified the path by rounding off its corners using arcs of a unit circle.

3.2. Strategy 2: J. Pach. According to the weak version of Problem 2, Pach wanted to show the existence of a short path between two points in a square packing with side lengths at most 1. He managed to prove the following.

Theorem 3 (J. Pach [P77]). *Given any packing in the plane with squares of side length at most 1, and two points S and T at a distance d , not contained in any square, there exists a path from S to T avoiding the squares of length $\ell \leq \frac{3}{2}d + 4\sqrt{d} + 1$.*

How did J. Pach show the existence of a short path in a square packing?

One of the most valuable applications of the definite integral gives a way to discuss the average value of a function. Take for example a unit square whose projection on the x axis runs from a to b . Assign to each $c \in [a, b]$ the function value $\ell(c)$ which is equal to the length of the shorter boundary arc of the square needed to connect the two boundary points whose x coordinate are equal to c . If the square is in axis parallel position one can verify (by visualizing the “ Λ ” shaped graph of $\ell(x)$) that $\int_a^b \ell(x) = \frac{3}{2}$. Pach believed that the axis parallel position provides the largest value of this integral for every square position. Once he verified this, he was ready to explain how to connect two points, S and T , of distance $|ST| = d$ with a path of length $\frac{3}{2}d + o(d)$. Pach’s main idea was to consider a large rectangle where S and T were the midpoints of two opposite sides. Then starting at each point of the side containing S , using the evade a square and return to the line algorithm along a perpendicular line, he constructed a path reaching the opposite side. Now with his lemma concerning a single square, he was able to conclude that the integral of the path lengths along the the side containing S is at most $\frac{3}{2}$ of the area of the rectangle. Thus, somewhere there is a connecting path of length $\frac{3}{2}d$. By choosing the dimensions of the rectangle properly, he then used the return to the line heuristic along his path to get from S to the side containing T , and then travel to T with an additional $O(\sqrt{d})$ term.

3.3. Strategy 3: G. Fejes Tóth. To proved the strong version of Problem 2 G. Fejes Tóth was able to find the existence of a short path between points in a square packing with side lengths at most 1, with the following length.

Theorem 4 (G. Fejes Tóth [GFT78]). *Given any packing in the plane with squares of side length at most 1, and two points S and T at a distance d , not contained in any square, there exists a path from S to T avoiding the squares of length $\ell \leq \frac{3}{2}d + \frac{1}{2}$.*

G. Fejes Tóth's proof is based on an innocent elementary lemma, which later was also applied by other authors.

Lemma 1 (G. Fejes Tóth [GFT78]). *If one travels along a line e and hits a square at point X , then they can travel to a vertex V along the boundary of the square so that the path length $\ell(X, V)$, from X to V , is at most $\frac{3}{2}$ times the length of the projection of vector $\overrightarrow{XV}$ onto line e .*

How did G. Fejes Tóth show the existence of a short path in a square packing?

The above lemma implies, that if a direction $\vec{V}$ is set, then starting at S one can make a path so that i) between squares the path contains segments parallel to $\vec{V}$, ii) when the path evades squares, then it makes a significant (a factor of $\frac{2}{3}$) progress towards the target relative to the length of the path shared with the squares. The obvious problem is that such a path may not go through the target point, even if one changes the direction continuously. Tóth's has an elegant way to overcome this difficulty. First he finds a direction so that at encountering a square, the efficiency while proceeding left or right to a vertex is equal. In words, he creates a two-pronged fork, both prongs being efficient. Moreover he does this so that the target point T belongs to the region between the two prongs of the fork. He still does not have a connecting path, but at this point he starts an equally efficient path from opposite direction than $\vec{V}$. This third path eventually has to intersect one of the two prongs, completing an efficient connecting path.

3.4. Strategy 4: C. Papadimitriou and M. Yannakakis. Papadimitriou and Yannakakis in 1991 were the first to find an algorithmic proof for the weak version of Problem 2. They developed the *bias heuristic*, which will be described shortly, to prove the following.

Theorem 5 (C. Papadimitriou and M. Yannakakis [PY91]). *Given any packing in the plane with squares of side length at most 1, and two points S and T at a distance d , not contained in any square, the bias heuristic gives a path from S to T avoiding the squares of length $\ell \leq \frac{3}{2}d + O(d^{2/3})$.*

How did C. Papadimitriou and M. Yannakakis use the bias heuristic to construct a short path in a square packing?

The authors first considered squares which were all parallel to the horizontal line ST . It was natural to go horizontally and when a square was hit go vertically to the nearest corner and proceed horizontally. Unfortunately, this strategy may lead a person far from the horizontal line ST . At this point Papadimitriou and Yannakakis introduced a bias function which updates after each encountered square. It changed incrementally by $\frac{1}{\sqrt{d}}$. When faced with a square, the traveler had a bias towards the corner closer to the line ST . Every time the traveler moved away from ST the bias function got increased, otherwise decreased by one increment. This way, as the traveler moves further from the line ST , successive squares become more and more likely to favor traveling back towards the line ST even if it comes at some cost. They were then able, with careful analysis, prove the desired upper bound. The difficulty came after

this point. First they generalized this idea for packings, where the squares were still parallel, but could have any edge length ≤ 1 . See Fig. 3(c) for a visual of the bias. The surprising part was that this heuristic could be applied to the general case, where the squares could have arbitrary orientation. The basic idea was to discretize the slopes of the squares, make groups of those squares whose edge slope was almost the same. And apply the last bias strategy for each group independently from the other groups.

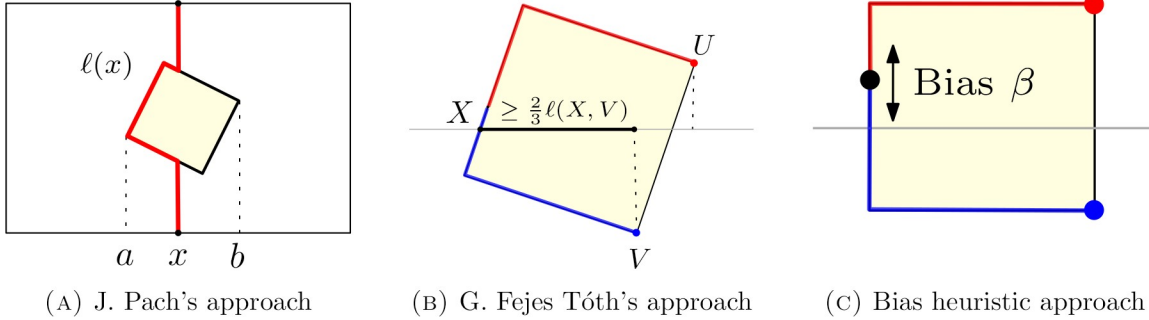


FIGURE 3. Three approaches for finding short paths in square packings detailed in Section 3.

3.5. Strategy 5: A. Bezdek and O. Henderschedt. The authors revisited the bias algorithm of C. Papadimitriou and M. Yannakakis. Their goal was to develop a much simpler algorithm in terms of both implementation and verification. By implementing the *evade a square and go straight towards the target heuristic*, which will be described shortly, they were able to reduce the validation of correctness for the weak version of Problem 2 from 9 pages to 1 page (see the next section for details). In this paper they proved the following.

Theorem 6. *Given any packing in the plane with squares of side length at most 1, and two points S and T at a distance d , not contained in any square, the evade a square and go straight towards the target heuristic gives a path from S to T avoiding the squares of length $\ell \leq \frac{3}{2}d + O(\sqrt{d})$.*

How did A. Bezdek and O. Henderschedt use the go straight towards the target heuristic to construct a short path in a square packing?

When dogs are running to fetch a toy, they never lose sight of the target and they always run directly towards the target. When faced with obstacles, they do the minimum detour to get around them, but once the field is clear in front of them, they reset the running direction and continue running directly towards the target. Both authors have dogs, admire their intelligence, and decided to prove that in fact dogs use the optimal strategy getting to the target. They could not see a pattern on how their dogs would go around a square obstacle (left or right), this is the point where they had to be creative and find a precise instruction/algorithm suitable for a simple elementary proof.

4. DETAILED ANALYSIS OF THE ‘EVADE A SQUARE AND GO STRAIGHT TOWARD THE TARGET HEURISTIC’

In this section we explicitly state and give the new proof the *go straight towards the target heuristic* indeed generates a connecting path of length $\leq \frac{3}{2}d + o(d)$.

Evade a square and go straight towards the target heuristic

- (1) (Initialization) Set direction $\vec{d}$ to be equal to the start-target direction $(\vec{ST})$. Denote the distance $|ST|$ by d and start the path at S .
- (2) Travel along direction $\vec{d}$ towards T until T is reached (in this case stop) or a square $Q = Q_1Q_2Q_3Q_4$ is hit at a point X . Assume $X \in Q_1Q_2$. Denote by Y the point where the line of direction $\vec{d}$ through T leaves square Q .
- (3) **If** the distance $|XT| > 2\sqrt{d}$, then
 - If** $Y \in Q_3Q_4$, then travel along the boundary of square Q to that endpoint of the edge Q_3Q_4 for which the gain along direction $\vec{d}$ is at least $\frac{2}{3}\ell(X, Q_i)$. Note that a Lemma (exact number needed) of G. Fejes Tóth shows that one of the two end points always has this property. Then reset $\vec{d}$ to be equal to $\vec{Q_iT}$ and go to (2).
 - Else** X and Y belong to consecutive edges of Q . Connect X and Y , along these two edges. Then go to (2).
 - Else** Connect X to T using the simple *return to the line XT heuristic*.
- (4) **Stop**

Theorem 7. Let $\mathcal{P}$ be an arrangement of non-overlapping squares of edge lengths ≤ 1 and let S and T be two points at distance d from each other, not contained in any square of $\mathcal{P}$. The **Evade a square and go straight towards the target heuristic** yields a path $\ell(S, T)$ from S to T avoiding every square of $\mathcal{P}$ such that $\ell(S, T) \leq \frac{3}{2}d + O(\sqrt{d})$.

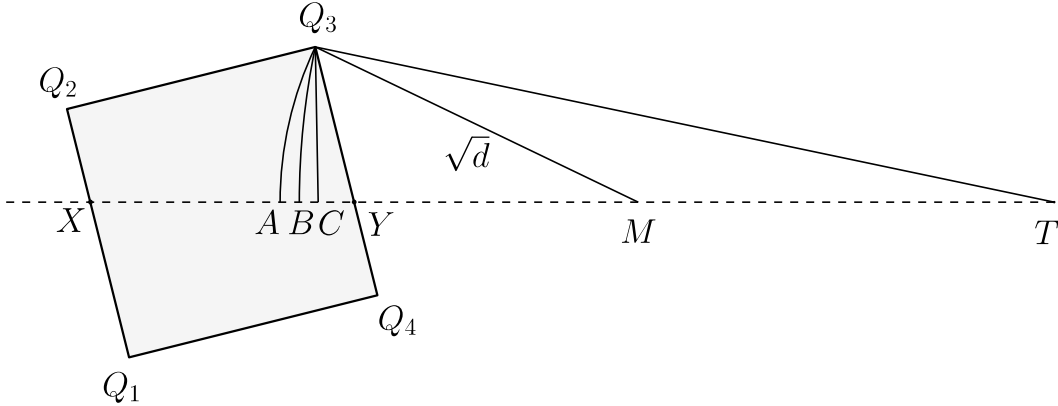


FIGURE 4. Proof in case of Step (3)/if/if in Theorem 7.

Proof. First we consider the special case when all obstacles are unit squares. It turns out that the general case is proved exactly in the same way at the price of including scaling factors in the inequalities. We use all the notation introduced in the above algorithm. We

will study those pieces of the path which are produced by steps (3)/if/if, (3)/if/else and (3)/else of the above algorithm and give upper bounds for the length of the pieces in the form of a constant times the progress made towards T .

Case of Step (3)/if/if:

Let $Q_1Q_2Q_3Q_4$ be consecutive vertices of square Q so that $X \in Q_1Q_2$ and $Y \in Q_3Q_4$. Without loss of generality assume Q_3 is the vertex chosen by Lemma (exact number needed) of G. Fejes Tóth. Let C_T be the circle centered at T with radius TQ_3 . Let C_M be the circle centered at $M \in YT$ with radius $|MQ_3|$ equal to $\sqrt{d}$.

Let A be the closest point of circle C_M to X , let B be the closest point of circle C_T to X and let C be the projection of Q_3 onto the line XT . Clearly $|XA| \leq |XB| \leq |XC|$ (See Fig. 4).

We already said $1 \leq \ell(X, Q_i) \leq \frac{3}{2}|XC|$. Thus $|XC| \geq \frac{2}{3}$.

Since $|Q_3C| \leq 1$, by the Pythagorean theorem,

$$|AC| = |AM| - |CM| \leq \sqrt{d} - \sqrt{d-1} < \frac{1}{2\sqrt{d-1}}$$

.

The last two lines imply that for $d > 3$ we have $|XB| > |XA| > \frac{1}{3}$.

By Tóth's lemma we have

$$\ell(X, Q_i) \leq \frac{3}{2}|XC| < \frac{3}{2}(|XB| + |AC|) < \frac{3}{2} \left(|XB| + \frac{1}{2\sqrt{d-1}} \right)$$

.

Replacing the numerator 1 in the last fraction with $3|XB|$ we get

$$\ell(X, Q_i) < \frac{3}{2} \left(|XB| + \frac{3|XB|}{2\sqrt{d-1}} \right) \quad (*)$$

.

Case of Step (3)/if/else:

Consider the shorter boundary arc XY along the perimeter of a single unit square. If X and Y belong to opposite edges then this arc has length ≤ 2 while the distance $|XY| \geq 1$, thus $2|XY|$ is an upper bound for the connecting path. If X and Y belong to adjacent edges then $\sqrt{2}|XY|$ is an upper bound for the connecting path. Thus the total path length in Step (3)/if/else has length

$$\ell(X, T) \leq 2|XT| \leq 4\sqrt{d}. \quad (**)$$

.

Case of Step (3)/else:

It is easy to see that connecting X to Y using adjacent edges of a square makes the the least progress relative to the path length, when X and Y are diagonally opposite vertices of a square. Thus Step (3)/else produces a path for which

$$\ell(X, Y) \leq \sqrt{2}|XY| \leq \frac{3}{2}|XY|. \quad (***)$$

.

Finally, note that in (*) the length $|XB|$, in (**) the length $|XT|$ and in (***) the length $|XY|$ are the true progresses, thus summing up those for each steps of the algorithm we get

$$\ell(S, T) \leq \frac{3}{2}d + o(d).$$

□

5. CONNECTING TWO POINTS WITH A PATH ALONG THE BOUNDARY OF THE REGIONS OF A COVERING

Short path problems in planar coverings gained attention when G. Fejes Tóth asked Problem 3 concerning circle coverings. The status of this problem was described after the formal statement in Section 2. It is expected that similarly to the packing problems, short path problems in coverings will be considered for coverings with squares, parallelograms and later with constant width discs. In this section we give two bounds for the path lengths in a special family of square coverings.

Theorem 8. *Consider a covering of the plane by translates of a unit square. Assume that points S, T of distance d belong to the boundary of two of the squares. We show that if the segment ST is parallel to one of the edges of the squares, then*

- i) *there is a covering, where S and T cannot be connected with a path contained along the edges of the squares of length $\ell < 2d$,*
- ii) *in every covering, S and T can be connected along the boundary of the squares, with a path length $\ell \leq \frac{8}{3}d + 8$.*

Proof of Theorem 8i. Consider the following covering. Stack unit squares to form a vertical column, infinite at both ends. Make a layer (blue shaded squares on Figure 5) with horizontal translates of this column so that the columns are separated by a distance ϵ from each other. Make a second layer (transparent squares) by translating down the first layer by a distance $\frac{1}{2}$ and horizontally to right by a distance $\frac{1+\epsilon}{2}$ to get a square covering.

Now choose points S and T so that they lie on a long horizontal segment and consider any path connecting S and T along the edges of the squares. Subdivide this path with points of the ϵ long segments between the columns of the first layer. Notice that the path between consecutive subdivision points has to have at least two vertical segments of length $\frac{1}{2}$. Thus, in order to gain horizontal distance of $1 + \epsilon$ one has to travel vertically a total distance of 1. □

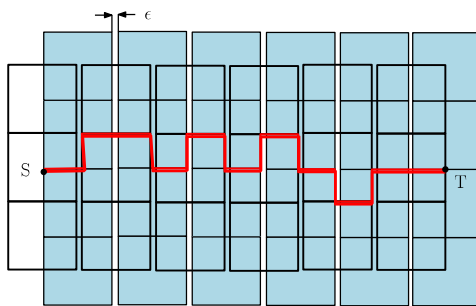


FIGURE 5. Square coverings and points S and T so that all connecting path along the edges of the squares have length $\geq 2|ST|$.

Proof of Theorem 8ii. First we will introduce some terminology and notation. The actual proof will be divided in two parts. The first half of the proof will be called the reductions part. This part will allow us to consider only special coverings which are worst in the sense of finding short connecting path. The second half of the proof contains the construction of the path together with a length estimate.

Part 1: Terminology and notation. We will call a covering of a segment by squares *minimal* if removing any of the squares makes the segment no longer completely covered. A minimal cover of the segment ST is called ϵ -compressed if every second square, when the squares are listed from S to T , are at distance $< \epsilon$ from each other. A path will be called *monotone* if traversing along the path from S to T , the projection of the moving point on the segment ST moves monotonically towards T .

From here on we consider only minimal covers of segment ST .

The following assumption and notation will help us in explaining the proof. All the unit squares of the minimal cover of ST have a pair of horizontal edges. Thus the segment ST is also horizontal with left endpoint at S . The squares of $s_1, s_2, \dots, s_n$ are indexed from left to right. Let $x_1 = S$, and for $i = 2, \dots, n$, let x_i be a common point of s_{i-1} and the left vertical edge of s_i . Notice that the square s_{n+1} is not part of the minimal cover.

Part 2: Reductions, worst paths. We will make two reductions.

1st reduction: In the rest of the proof we assume X_1 and X_n belongs to the left vertical edges of squares s_1 and s_n respectively. If X_1 does not belong to the left vertical edges of squares s_1 , then first we use a path of length ≤ 2 to get to x_2 and will relabel x_2 as S (the case at X_n is handled similarly). In other words, this reduction involved relocating S and T to the proper edge at the cost of the path length, in the worst case, adds a constant $2 \cdot 2$. Also note that after relocating S and T the distance between them decrease. The overall effect of these changes to the inequality in ii) will be a constant 4.

2nd reduction: First we prove a lemma, which reduces the short path problem to a square covering where every second neighbour of the squares are almost tangent to each other:

Lemma 2. *Assume that a minimal square covering satisfies the first reduction. We show that the ‘worst’ case for proving an upper for the path length connecting S and T , is when the square covering is ϵ -compressed.*

Proof of Lemma 2. Start with a minimal cover and with a monotone path P along the edges. We will shift continuously the squares to the left without creating triple overlaps. to show how this is done, we provide the following routine:

- i:** Fix the position the first two squares of the minimal cover.
- ii:** Shift the union of the remaining squares to the left until the third square gets ϵ -close to the first one or the 4th square gets ϵ -close to the second one. If the later happens first, then follow by shifting the third to the left until it also gets ϵ close to the second square. Fix the position of the first three (or first 4 squares resp.)
- iii:** Keep repeating ii) with the understanding the last two of the fixed squares plays the role of the first two square.

This will shrink some of the horizontal segments of the path P , but not the vertical ones. The vertical projection of the path will decrease by the exact same length. Since

the function $f(x) = \frac{a+x}{b+x}$ is a decreasing function for $a > b$, we have that the ratio $\frac{L(P)}{d(S,T)}$ increases. ■

Part 3: Construction of the paths and length estimates

Consider now a minimal, ϵ -compressed cover of ST so that S and T belong to the left vertical edges of squares. Also assume that the squares are indexed from left to right from 1 to n . We will group the squares into triplets and according to the following lemma we will show how to travel through the triplets efficiently. The following lemma will explain what is considered efficient and the paragraph after the proof of this lemma will explain the overall upper bound of the path.

Lemma 3. *Let A, B, C be three consecutive unit squares of an ϵ -compressed minimal cover of the horizontal line l . Let U be a random point of the left vertical edge of square A , and let VW be the right edge of square C where vertex V is above vertex W . Then, U can be connected to both V and W with a path of length $\leq 4 + \epsilon$ along the boundary edges of the squares without using the edge VW .*

Proof of Lemma 3: We distinguish two cases (Figure 6).

Case 1: The vertical positions of the squares A, B, C in this order do *not* increase or decrease monotonically.

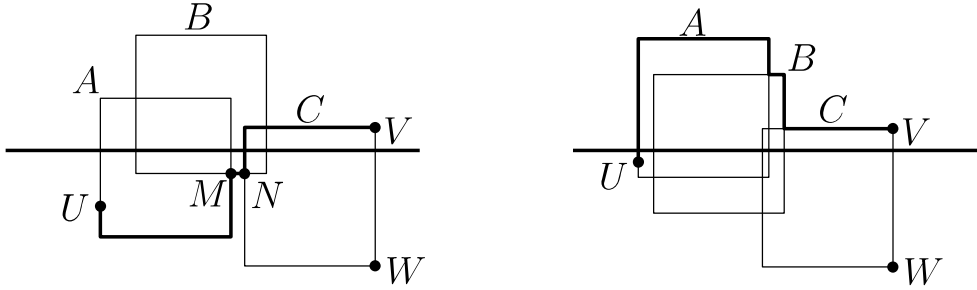
In this case the middle square B has a horizontal edge, which intersects one of the vertical edges of both squares A and C . Denote these intersection points from left to right by M and N . Since the squares are ϵ -compressed, we have $MN \leq \epsilon$. Let us connect U to M with a path length ≤ 2 using the shorter boundary arc around square A . Finally, let us connect N to vertex V (vertex W resp.) with a path length ≤ 2 using the shorter boundary arc around square C . Here the shorter path does not include the edge VW , thus the combined path has length $\leq 4 + \epsilon$.

Case 2: The vertical positions of the squares A, B, C in this order monotonically increase or decrease.

If we want to connect U to V , then first connect U to the upper left vertex of square A along the left vertical edge of A with a segment of length ≤ 1 . Then proceed along the upper boundary arc of the union of the three squares to vertex V . The horizontal segments of this later path has length $\leq 2 + \epsilon$, while the vertical segments of this path have a total length ≤ 1 . Thus, the combined path has length $\leq 4 + \epsilon$.

If we want to connect U to W , then we connect U to the lower left vertex of square A and then continue along the lower boundary arc of the union of the three squares to vertex W . ■

Finishing the proof of Theorem 8ii. We group the squares in the ϵ -reduced minimal cover into triplets and for each triplet we construct a path connecting x_{3i} to x_{3i+2} with a path length ≤ 4 . Since the number of square triplets is integer part $\frac{n}{3}$. If two squares are left without triplets we can finish the path with adding a piece of path of length ≤ 4 . Since the squares with even label (and with odd label resp.) are disjoint we have that $n \leq 2d$, thus we have a path of length $\leq 4\frac{2d}{3} + 4$. In the limit this gives the upper bound of $\ell\frac{8}{3}d + 4$. If the first reduction was needed we get the overall upper bound for the length of the connecting path $\frac{8}{3}d + 8$. □

FIGURE 6. The paths between U and V in Lemma 3.

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